

I. Project Title

ArtGuard AI: A Transparency Tool for Identifying AI-Generated Artwork

II. Problem Statement

With the endless amount of generative AI tools available, the art community frequently face issues regarding copyright transparency, uncredited AI-generated images, and the non-consensual use of original art styles. A large portion of art spaces online struggle with verifying whether a piece was handcrafted by an actual person or generated by AI. This problem creates trust issues in the community and affects aspiring artists trying to build authentic portfolios. Developing a lightweight tool to flag potential AI-generated traits helps maintain transparency and protect creative integrity.

III. Project Objectives

1. **Detect Generative Patterns:** Identify common visual patterns that are typically in AI-generated images.
2. **Provide Transparency Scores:** Generate a clear percentage score estimating the likelihood that an image is human-made vs. AI-generated.
3. **Promote Responsible AI use:** Help creators clearly disclose AI assistance in their artistic workflows.

IV. Planned Features

- **Image Upload & Analysis:** Allows users to drag and drop an image file for immediate analysis.
- **Metadata & Artifact Inspector:** Checks embedded EXIF data, pixel noise consistency, and symmetry anomalies
- **Authenticity Report Generator:** Display a breakdown score (ex. "87% High Probability of AI Generation") along with detected visual cues.

V. Planned Inputs and Outputs

- **User Inputs:**
 - Image file (PNG or JPEG)
 - Optional artist notes or self-disclosure tags (ex. "Human Artwork," "AI-Assisted," "Fully AI-Generated")

Computer Science 2: Project Proposal (Initial Draft)

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- **System Outputs:**

- Likelihood score for AI generation (percentage)
- Visual highlight map showing flagged areas
- Summary report downloadable as a text file

VI. Logic Plan

A. Pseudocode

START

DISPLAY "Welcome to ArtGuard AI Inspector"

INPUT user_image_file

IF user_image_file IS NULL OR INVALID_FORMAT THEN

DISPLAY "Error: Please upload a valid PNG or JPEG image."

EXIT

ENDIF

metadata_score = CHECK_METADATA(user_image_file)

artifact_score = CHECK_PIXEL_ARTIFACTS(user_image_file)

total_ai_probability = (metadata_score * 0.4) + (artifact_score * 0.6)

DISPLAY "Analysis Complete."

DISPLAY "Estimated AI Probability: " + total_ai_probability + "%"

IF total_ai_probability > 70 THEN

DISPLAY "Classification: High Probability of AI Generation"

ELSE IF total_ai_probability > 30 THEN

DISPLAY "Classification: Mixed / Potential AI Assistance"

ELSE

DISPLAY "Classification: Likely Human-Created"

ENDIF

END